

Software Requirements Specification

StyleHouse — Online Clothing Store

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for **StyleHouse**, an e-commerce website for a clothing brand. It is intended to serve as the primary reference for the design, development, testing, and deployment of the platform. The document is written for the development team, quality assurance engineers, UI/UX designers, project managers, and business stakeholders involved in building the product.

1.2 Scope

StyleHouse is a web-based (and future mobile-ready) e-commerce platform that allows the clothing brand to sell apparel and accessories directly to consumers. The system will allow customers to browse products by category, size, color, and collection; search and filter products; manage a shopping cart and wishlist; complete secure checkout with multiple payment options; track orders; and manage their account and addresses. The platform will also provide an administrative back-office for managing products, inventory, orders, discounts/coupons, customers, and content. The scope of this document covers the customer-facing storefront, the admin panel, and the supporting APIs, but excludes third-party logistics/warehouse management systems, which are treated as external integrations.

1.3 Intended Audience

- **Development Team** — to implement the frontend, backend, and database as specified.
- **QA/Testing Team** — to design test cases and validate acceptance criteria.
- **UI/UX Designers** — to design wireframes and interfaces consistent with requirements.
- **Project Managers** — to plan sprints, milestones, and resource allocation.
- **Business Stakeholders** — to review and approve scope, features, and priorities.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
SKU	Stock Keeping Unit — unique identifier for each product variant
UI/UX	User Interface / User Experience
API	Application Programming Interface
COD	Cash on Delivery
JWT	JSON Web Token, used for authentication
CMS	Content Management System
PCI-DSS	Payment Card Industry Data Security Standard
Admin	A back-office user who manages catalog, orders, and site content
Guest User	A visitor browsing/purchasing without creating an account

1.5 References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications
- OWASP Top 10 — Web Application Security Risks
- PCI-DSS Compliance Guidelines for Online Payment Processing
- WCAG 2.1 — Web Content Accessibility Guidelines

1.6 Overview

The remainder of this document is organized as follows: Section 2 provides an overall description of the product, its users, and constraints. Section 3 details specific functional requirements organized by module. Section 4 describes external interface requirements. Section 5 covers non-functional requirements. Section 6 describes the data model at a high level, and Section 7 lists assumptions, dependencies, and future scope.

2. Overall Description

2.1 Product Perspective

StyleHouse is a new, self-contained e-commerce product built for a clothing brand that currently may sell through offline retail or a basic informational website. The platform will consist of three primary components: a **customer-facing storefront** (responsive web application), an **admin dashboard** for internal operations, and a **backend API layer** connecting the storefront, admin panel, database, and third-party services (payment gateway, shipping provider, email/SMS notification service).

2.2 Product Functions

At a high level, the system shall provide the following major functions:

- User registration, login, and profile/address management
- Product catalog browsing with categories, filters, and search
- Product detail pages with size/color variants, images, and stock status
- Shopping cart and wishlist management
- Secure checkout with multiple payment methods and order placement
- Order tracking, order history, cancellations, and returns/refunds
- Coupon codes, discounts, and promotional banners
- Product reviews and ratings
- Admin catalog, inventory, order, and customer management
- Email/SMS notifications for order and account events

2.3 User Classes and Characteristics

User Class	Description
Guest User	Browses products, searches/filters, adds to cart; must register or log in to complete checkout.
Registered Customer	Full access: checkout, order history, wishlist, saved addresses, reviews, profile management.
Admin / Store Manager	Manages products, inventory, categories, pricing, and promotions via the admin dashboard.
Order/Support Staff	Views and updates order status, processes returns/refunds, responds to customer queries.
Super Admin	Full system access including staff account management, site configuration, and reporting.

2.4 Operating Environment

- **Client side:** Modern web browsers (Chrome, Safari, Firefox, Edge) on desktop, tablet, and mobile screen sizes.

- **Server side:** Cloud-hosted backend (e.g., AWS/GCP/Azure) with a RESTful or GraphQL API layer.
- **Database:** Relational database (e.g., PostgreSQL/MySQL) for transactional data; optional NoSQL/cache layer (e.g., Redis) for sessions and catalog caching.
- **Third-party integrations:** Payment gateway (e.g., Razorpay/Stripe), shipping/courier API, email service (e.g., SendGrid), and SMS gateway.

2.5 Design and Implementation Constraints

- The platform must comply with PCI-DSS guidelines; card data shall never be stored on StyleHouse servers directly — payment gateway tokenization must be used.
- The system must be responsive and mobile-first, since a majority of fashion e-commerce traffic originates from mobile devices.
- The system must support horizontal scaling to handle seasonal sale traffic spikes (e.g., end-of-season sales).
- All customer-facing pages must meet WCAG 2.1 AA accessibility standards where feasible.

2.6 Assumptions and Dependencies

- It is assumed that a third-party payment gateway and shipping/courier partner will be available and integrated via API.
- It is assumed that product photography and catalog content will be supplied by the brand's marketing/design team.
- The system depends on reliable internet connectivity for real-time inventory and payment processing.
- Tax calculation rules (e.g., GST/VAT) are assumed to follow the brand's operating region and may require a dedicated tax-calculation service.

3. Specific Requirements

This section lists functional requirements grouped by module. Each requirement is uniquely identified (e.g., FR-AUTH-01) and assigned a priority: **High** (must-have for MVP), **Medium** (important, can follow shortly after MVP), or **Low** (nice-to-have / future release).

3.1 User Registration & Authentication

ID	Requirement Description	Priority
FR-AUTH-01	The system shall allow a new user to register using name, email, and password, or via a phone number with OTP verification.	High
FR-AUTH-02	The system shall allow login via registered email/phone and password, and support 'forgot password' via email/SMS reset link or OTP.	High
FR-AUTH-03	The system shall support social login (Google/Facebook) as an alternative sign-in method.	Medium
FR-AUTH-04	The system shall allow guest checkout without requiring account creation.	High
FR-AUTH-05	The system shall allow a logged-in user to update profile details, change password, and manage saved addresses.	High
FR-AUTH-06	The system shall log out a user and invalidate the session/token upon logout or token expiry.	High

3.2 Product Catalog & Browsing

ID	Requirement Description	Priority
FR-CAT-01	The system shall display products organized by categories (e.g., Men, Women, Kids, Accessories) and sub-categories (e.g., Shirts, Dresses, Footwear).	High
FR-CAT-02	Each product listing shall show a thumbnail image, name, price, discount (if any), and available rating summary.	High
FR-CAT-03	The system shall provide a product detail page with multiple images, description, fabric/material info, size chart, available sizes/colors, price, and stock availability.	High
FR-CAT-04	The system shall allow users to filter products by size, color, price range, brand/collection, discount, and customer rating.	High
FR-CAT-05	The system shall allow users to sort product listings by price (low-high/high-low), newest arrivals, and popularity.	Medium
FR-CAT-06	The system shall provide a keyword search with autosuggestions for products, categories, and collections.	High
FR-CAT-07	The system shall display 'Out of Stock' status and disable purchase for unavailable size/color variants.	High
FR-CAT-08	The system shall display related/recommended products on the product detail page (e.g., 'You may also like').	Low

3.3 Shopping Cart & Wishlist

ID	Requirement Description	Priority
FR-CART-01	The system shall allow users to add a product (with selected size/color and quantity) to the shopping cart.	High
FR-CART-02	The system shall allow users to update quantity or remove items from the cart, with real-time price recalculation.	High
FR-CART-03	The system shall persist the cart across sessions for logged-in users, and use local/browser storage for guests.	Medium
FR-CART-04	The system shall display cart subtotal, applicable discounts, shipping estimate, and final total before checkout.	High
FR-CART-05	The system shall allow users to add/remove products to/from a wishlist and move wishlist items into the cart.	Medium
FR-CART-06	The system shall validate stock availability at the time of checkout and notify the user of any changes since items were added.	High

3.4 Checkout, Payment & Orders

ID	Requirement Description	Priority
FR-CHK-01	The system shall allow users to select or add a shipping address during checkout, including address validation for required fields.	High
FR-CHK-02	The system shall allow users to select a shipping method (e.g., standard, express) with corresponding cost and estimated delivery date.	Medium
FR-CHK-03	The system shall support multiple payment methods: credit/debit card, UPI, net banking, digital wallets, and Cash on Delivery (COD) where enabled.	High
FR-CHK-04	The system shall integrate with a third-party payment gateway for secure payment processing and shall not store raw card details.	High
FR-CHK-05	The system shall allow users to apply a coupon/promo code at checkout, validating eligibility and applying the discount to the order total.	Medium
FR-CHK-06	The system shall generate a unique order confirmation with an order ID upon successful payment, and send a confirmation email/SMS.	High
FR-CHK-07	The system shall allow users to view order history and track the current status of each order (e.g., Placed, Packed, Shipped, Out for Delivery, Delivered).	High
FR-CHK-08	The system shall allow users to cancel an order prior to shipment, and request a return/exchange within a defined policy window after delivery.	High
FR-CHK-09	The system shall process refunds to the original payment method (or store credit) upon approved return/cancellation.	Medium

3.5 Reviews & Ratings

ID	Requirement Description	Priority
FR-REV-01	The system shall allow customers who purchased a product to submit a star rating (1-5) and a written review.	Medium
FR-REV-02	The system shall display the average rating and total review count on product listing and detail pages.	Medium
FR-REV-03	The system shall allow admins to moderate (approve/reject/remove) submitted reviews.	Low

3.6 Notifications

ID	Requirement Description	Priority
FR-NOT-01	The system shall send email/SMS notifications for account creation, order confirmation, shipping updates, and delivery.	High
FR-NOT-02	The system shall notify users of price drops or restocks for wishlisted items.	Low
FR-NOT-03	The system shall send promotional notifications (sales, new arrivals) to opted-in users.	Low

3.7 Admin Panel

ID	Requirement Description	Priority
FR-ADM-01	The system shall allow admins to create, edit, and deactivate products, including images, variants (size/color), price, and stock quantity.	High
FR-ADM-02	The system shall allow admins to manage categories and collections, and feature specific products/banners on the homepage.	High
FR-ADM-03	The system shall allow admins to view, update status, and manage all customer orders, including cancellations and returns.	High
FR-ADM-04	The system shall allow admins to create and manage discount coupons (percentage, flat amount, minimum order value, validity period).	Medium
FR-ADM-05	The system shall allow admins to view customer accounts and order history for support purposes.	Medium
FR-ADM-06	The system shall provide sales and inventory reports/dashboards (e.g., revenue, best-selling products, low-stock alerts).	Medium
FR-ADM-07	The system shall support role-based access control distinguishing Super Admin, Store Manager, and Support Staff permissions.	Medium

4. External Interface Requirements

4.1 User Interfaces

- A responsive storefront supporting desktop, tablet, and mobile breakpoints, following the brand's visual identity (typography, color palette, logo usage).
- A clean, distraction-free checkout flow (cart → address → shipping → payment → confirmation).
- An admin dashboard with sidebar navigation for Products, Orders, Customers, Discounts, and Reports.

4.2 Hardware Interfaces

No specialized hardware is required. The system shall be accessible via standard consumer devices (desktops, laptops, tablets, smartphones) with an internet connection.

4.3 Software Interfaces

Interface	Description
Payment Gateway API	Processes online payments (cards, UPI, wallets, net banking) and returns transaction status.
Shipping/Courier API	Generates shipment labels, retrieves tracking status, and calculates shipping rates.
Email/SMS Service	Sends transactional emails/SMS (order confirmation, OTP, shipping updates).
Tax/Invoice Service	Calculates applicable taxes (e.g., GST) and generates invoices per order.
Analytics Service	Tracks user behavior (e.g., Google Analytics) for marketing and UX insights.

4.4 Communication Interfaces

All client-server communication shall occur over HTTPS (TLS 1.2 or higher). The backend shall expose a RESTful (or GraphQL) API consumed by the web storefront and admin panel, using JSON as the primary data interchange format.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Product listing and search pages shall load within 2 seconds under normal load (up to 500 concurrent users).
NFR-02	Scalability	The system shall support horizontal scaling to handle at least 10x traffic during flash sales/seasonal events.
NFR-03	Security	The system shall enforce HTTPS, hash and salt stored passwords, use JWT/session tokens with expiry, and comply with PCI-DSS for payment handling.
NFR-04	Availability	The system shall maintain 99.5% uptime, excluding scheduled maintenance windows.
NFR-05	Usability	The storefront shall be intuitive enough for a first-time user to complete a purchase within 5 steps or fewer.
NFR-06	Accessibility	Key customer-facing pages shall meet WCAG 2.1 AA guidelines (alt text, keyboard navigation, contrast ratios).
NFR-07	Maintainability	The codebase shall follow modular architecture to allow independent updates to catalog, cart, and payment modules.
NFR-08	Data Backup & Recovery	The system shall perform automated daily database backups with a recovery point objective (RPO) of 24 hours.
NFR-09	Browser Compatibility	The storefront shall function correctly on the latest two major versions of Chrome, Safari, Firefox, and Edge.
NFR-10	Localization	The system shall support multi-currency and multi-language display as a future-ready capability, even if only one locale is enabled at launch.

6. High-Level Data Model

The following core entities and their key attributes describe the primary data model underlying the platform. Detailed schema design (data types, indexes, relationships) will be handled in a separate database design document.

Entity	Key Attributes
User	user_id, name, email, phone, password_hash, role, created_at
Address	address_id, user_id, label, street, city, state, postal_code, country, is_default
Product	product_id, name, description, category_id, brand, base_price, images, status
ProductVariant	variant_id, product_id, size, color, SKU, stock_quantity, price_override
Category	category_id, name, parent_category_id
Cart / CartItem	cart_id, user_id/session_id, variant_id, quantity, added_at
Order	order_id, user_id, status, subtotal, discount, shipping_fee, tax, total, placed_at
OrderItem	order_item_id, order_id, variant_id, quantity, unit_price
Payment	payment_id, order_id, method, gateway_txn_id, status, amount, paid_at
Coupon	coupon_id, code, discount_type, value, min_order_value, valid_from, valid_to
Review	review_id, product_id, user_id, rating, comment, created_at, status

7. Assumptions, Dependencies & Future Scope

7.1 Assumptions & Dependencies

- The brand will provide product catalog data, pricing, and imagery ahead of each development phase.
- A payment gateway merchant account and shipping partner account will be set up and available for integration testing.
- Business rules for returns, exchanges, and refund windows will be finalized by the business team before checkout module development.

7.2 Out of Scope (Current Phase)

- Native mobile applications (iOS/Android) — the platform will be a responsive web application only in this phase.
- In-store POS (point-of-sale) integration.
- AI-based size/fit recommendation engine.

7.3 Future Scope

- Native mobile apps for iOS and Android.
- Loyalty/rewards program with points and membership tiers.
- AI-powered product recommendations and virtual try-on.
- Live chat / chatbot customer support integration.
- Multi-warehouse inventory management and vendor marketplace support.

8. Approval

Role	Signature	Date
Product Owner	_____	_____
Engineering Lead	_____	_____
QA Lead	_____	_____
Business Stakeholder	_____	_____